Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/18/2005
PI Name:	Theofanous, Theofanis G. Ph.D.		
Project Title:	The Limits of Coolability in Thermal Systems for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	theo@engineering.ucsb.edu	**Fax:**	FY 805 893-4927
PI Organization Type:	UNIVERSITY	**Phone:**	805 689-1311
Organization Name:	University of California, Santa Barbara		
PI Address 1:	Department of Chemical Engineering		
PI Address 2:	CRSS, 6740 Cortona Dr, Goleta CA 93117		
PI Web Page:	http://www.crss.ucsb.edu/		
City:	Santa Barbara	**State:**	CA
Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	10/01/2001	**End Date:**	11/30/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dinh, Truc-Nam (University of California Santa Barbara)		
Grant/Contract No.:	NAG3-2761		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is an integrative effort addressing practical aspects of boiling in the HEDS context. The work is a continuation and extension of a project that addressed coolability limits in pool boiling carried out over the previous funding cycle. A key result of this previous work was the identification of a scales separation phenomenon, that effectively "isolates" the microhydrodynamics on the heater surface from the chaotic two-phase flow motions away from it. This separation is effected by a persistent very high void fraction region (like a vapor "blanket"), and it is of immense significance in allowing to focus the physics of a long-standing seemingly intractable problem. Experiments and computations in this project will be used to examine applicability of scales separation as an organizing principle in convective boiling and boiling in fractional gravities. 1. Develop a basic understanding for the fine structure of the extended microlayer, as revealed through our "scale separation" idea and the dynamic thermal patterns on a nanofilm heater. 2. Define appropriate, measurable surface properties relevant to the rupture of microlayer, and relate those (quantitatively) to fluid properties and the rupture condition (the critical heat flux). 3. Determine the extent of applicability of scales separation in reduced gravity; that is, the point where CHF becomes water-supply-limited. 4. Determine the infleunce of convection on the results of Objetives 1 and 2. 5. Translate the results of all previous objectives to terms usable for predicting and enhancing CHF, as well as for use in a two-phase CFD framework. This project will provide a basis to develop a physics-based understanding of Boiling Crisis phenomenon in pool boiling (horizontal and vertical flat plates), and weakly-convecting flow boiling (vertical channel flow). We can then translate this understanding to a means for quantitative prediction of critical heat flux, as a function of local (near-wall) two-phase flow conditions, appropriate and convenient for interrogation (modelling) in the frame of detailed multiphase CFD simulations. Furthermore, the thus-gained understanding and experience can be used to enhance the coolability limits in both pool and forced convection flows by manipulating surface characteristics, and perhaps even flow geometries.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has been leveraged with a contract from the Department of Energy concerning the coolability limits of nuclear reactor vessels that in case of severe accidents are immersed in water pools by flooding the outside area with water. In turn the results of this fundamentally oriented work had a tremendous influence and impact on the large scale experiments we run for this nuclear application, and helped resolve this practical issue even though we are still in the process of developing a basic understanding.
Task Progress:	The purpose of this work is to understand the basic reason(s) that there is an upper limit on the level of power that can be removed from a heat-generating body by boiling of a liquid coolant. Such understanding is needed for reliable prediction, especially for systems as in space applications, for which a trial and error approach is not an option. Such understanding is also essential for realizing the full cooling potential of boiling, and thus opening up the way for innovative power production and conversion systems, as well as for addressing the cooling needs of new kinds of scientific equipment and instrumentation involving extremely intense heat sources (microelectronics, lasers, etc.). A wide range of terrestrial, aviation, and space technologies stand to benefit from such advancements. The symptoms of exceeding the cooling capacity of boiling are quite straightforward. The boiling surface dries out, and the temperature rises rapidly, thus leading to melting and destruction (hence the name “burnout”). Our problem is to find out why, in the presence of plentiful liquid around, the surface dries out. Research in the past half century led to the general concept, and belief, that the basic mechanism is insufficient liquid supply to the surface, this deficiency in supply being due to the repulsive effect of vapor generated at, and leaving the surface. This so-call hydrodynamic origin of burnout has led to several conceptual models that were translated to equations that purport to interpret experimental data. On the other hand, it is quite clear that understanding is still lacking, that predictions and control are not reliable, and that engineering system designs still rely on empirical correlations and trial and error approaches. In the first phase of our work under NASA support we demonstrated that in pool boiling the hydrodynamics set only the stage in a broad sense, rather than affecting coolability through some criticality condition as envisioned previously. This radical departure from the past is manifest in terms of the “scales separation phenomenon” discovered in this work. In the present, second phase, of the project the specific goal is to understand scales separation with all its implications for pool boiling, and to find out the extent to which it dominates behavior in convective (flow) boiling. Simply expressed, scales separation is due to a thin vapor layer that blankets the heater surface, while a micron-scale liquid film continues to provide cooling by evaporation. Dryout originates on this film as micron-scale spots, that spread and eventually merge to cause burnout. Prior to the inception of such an unstable condition, the thin film remains well organized, and its uniformity of behavior is in stark contrast to the chaotic liquid motions above it (above the vapor blanket). We say the heater-microlayer system is autonomous, while any excess liquid that reaches it, is rejected by the action of vapor production. In this way we have transformed and focused our problem (of coolability limits) to the stability of dry spots that form at the points at which vapor bubbles are nucleated on the heater surface. This in turn lead to hypotheses, and experiments that confirmed the importance of nanomorphology, and chemical constitution of the surface, as well as of the chemical composition of the liquid (including nanoparticles and impurities) being boiled, in affecting this onset of instability, and thus the power level needed for burnout. These experimental results in turn led to theoretical concepts involving microhydrodynamic behavior, such as molecular wicking (a mechanism for enhancing the spread of liquid molecules towards a dried-out region of the heating surface), and of vapor recoil (a mechanism that resists spreading). Still taking advantage of the scales separation phenomenon we are now working with the Microlayer itself, in a spray apparatus that allows independent control of thickness and detailed, high resolution observations from above (high speed video) as well as from bellow (high speed infrared visualization). In parallel we developed a forced convection apparatus that can conveniently accommodate the heater test sections from our pool boiling experiments, and we are entering initial testing in this too. This work has been leveraged with a contract from the Department of Energy concerning the coolability limits of

nuclear reactor vessels that in case of severe accidents are immersed in water pools by flooding the outside area with water. In turn the results of this fundamentally oriented work had a tremendous influence and impact on the large scale experiments we run for this nuclear application, and helped resolve this practical issue even though we are still in the process of developing a basic understanding.

BETA Progress

While the scales separation phenomenon has been known for as long as ~6 years (we first presented in 2000 and published in 2002), we have had great difficulties in fully elucidating what is the nature of the micro-hydrodynamics on the liquid film that covers, and cools, the heater at heat fluxes of over ~1 MW/m2. There several reasons for these difficulties. One is that that so far the film was accessible only through infrared imaging from bellow. While this was the means for discovering the phenomenon to begin with, much more was needed to interpret these data quantitatively as needed for revealing and understanding the full dynamics of the behavior—rapidly advancing and receding contact lines, bubble nucleation and growth, contact line instability. This was to be resolved by gaining optical access to the microlayer from above, so the second difficulty was to make robustly operational nanofilm heaters and learn how to irrigate them from above so as to reproduce the extended microlayer as it exists in pool boiling, but now in full view. In particular we had to eliminate edge effects, and be able to produce a stable film of any desired thickness down to ~10 microns. The third difficulty arose when we discovered, through visualization with a new high speed video camera (a Phantom 7, borrowed from one of our other projects) that the to resolve the micro-hydrodynamics we had to image at a minimum of 10 KHz. Thus we had to upgrade our IR camera, a costly but necessary task which we managed to complete about one month ago. A forth difficulty was found in modifying the software so as to allow simultaneous triggering and synchronization of the two cameras. Finally, a fifth problem to overcame was being able to take videos at 10 KHz, and in the presence of steaming corresponding to 1 to 2 MW/m2, without disturbing the film, and yet at very high magnification as needed to “see” microscopic features of the film and contact line motions.

Now that all finally worked we are able to answer a question that troubled us from the beginning of this discovery of scales separation—the bubble bases seen as cold spots on the infrared images were too big to be understood as such, if they were to be growing within a liquid layer that is only 10’s of microns in thickness. A publication is in preparation now.

At this time we can routinely age our heaters to achieve critical heat fluxes around 1.8 MW/m2, and we had one heater that went up to 2.9 MW/m2. Now we are in the intense production effort of taking and analyzing such synchronous data. Once this is brought to completion we will be ready to do the same in a flow boiling apparatus that is essentially complete.

Bibliography Type:	Description: (Last Updated: 10/17/2007)
Articles in Peer-reviewed Journals	R.R. Nourgaliev, T.N. Dinh, and T.G. Theofanous "A Pseudocompressibility Method for the Simulation of Multiphase Incompressible Flows" Intern. J. Multiphase Flows, Vol.30, pp.901-937. 2004 (available on line June 2004), Jun-2004 , Jun-2004
Articles in Peer-reviewed Journals	R.R. Nourgaliev, T.N. Dinh, T.G. Theofanous, and D.D. Joseph "The Lattice Bolztmann Equation Method: Theoretical Interpretation, Numerics, and Implications" Journal of Multiphase Flow, Vol.29 (1), pp.117-169, 2003. , Jan-2003 , Jan-2005
Papers from Meeting Proceedings	T.N. Dinh, J.P. Tu, A.T. Dinh, T.G. Theofanous "Nucleation Phenomena in Boiling on Nanoscopically Smooth Surfaces" 41st Aerospace Sciences Meeting (Thermophysics), Reno, Nevada, Jan 2003. AIAA-2003-1035. (AIAA 2003 Best Paper Award in Thermophysics) Jan-2003 Jan-2005
Papers from Meeting Proceedings	T.N. Dinh, J.P. Tu, A.T. Dinh, T.G. Theofanous "Nucleation Phenomena in Boiling on Nanoscopically Smooth Surfaces " 41st Aerospace Sciences Meeting (Thermophysics), Reno, Nevada, Jan 2003. AIAA-2003-1035. (AIAA 2003 Best Paper Award in Thermophysics) Jan-2005
Papers from Meeting Proceedings	T.N. Dinh, J.P. Tu and T.G. Theofanous "Hydrodynamic and Physico-Chemical Nature of Burnout in Pool Boiling" International Conference on Multiphase Flow, Yokohama, Japan, Paper 296. 14p. May-2004
Papers from Meeting Proceedings	J.P. Tu, T.N. Dinh and T.G. Theofanous "An Experimental Study of Boiling Heat Transfer with Nanoparticles" 6th International Symposium on Heat Transfer, Beijing Jun-2004
Papers from Meeting Proceedings	T.N. Dinh, Y.Z. Yang, J.P. Tu, R.R. Nourgaliev and T.G. Theofanous "Rayleigh-Bénard Natural Convection Heat Transfer: Pattern Formation, Complexity and Predictability" 2004 International Congress on Advances in Nuclear Power Plants, Pittsburgh, PA, Jun-2004
Papers from Meeting Proceedings	T.N. Dinh, J.P. Tu, Y.Z. Yang, R.R. Nourgaliev and T.G. Theofanous "Characterization and Predictability of Transient Heat Transfer in an Unstably Stratified Layer during Power Startup" 37th AIAA Thermophysics Conference, Portland, OR, June 27-30, 2004. AIAA-2004-2733. Jun-2004
Papers from Meeting Proceedings	T.N. Dinh, J.P. Tu, and T.G. Theofanous "Burnout in High Heat-Flux Boiling: Hydrodynamic and Physico-Chemical Factors" 42st Aerospace Sciences Meeting (Thermophysics), Reno, Nevada, Jan 2004. AIAA-2004-0165. Jan-2004
Papers from Meeting Proceedings	T.N. Dinh, J.P. Tu, T. Salmassi, T.G, Theofanous "Limits of Coolability in AP1000-Related ULPU-2400 Configuration V Facility" International Topical Meeting on Nuclear Reactor Thermal Hydraulics, Seoul, Korea, Oct, 2003 Oct-2003 Jan-2005

Papers from Meeting Proceedings	J.P., Tu, T.N. Dinh,, T.G. Theofanous "Enhancing Resistance to Burnout via Coolant Chemistry" 10th International Topical Meeting on Nuclear Reactor Thermal Hydraulics, Seoul, Korean, Oct, 2003 Oct-2003 Jan-2005